8.033 Fall 2025

Final Practice Problem Solutions

1 Short answers

(a) Precession of Mercury, gravitational lensing, clocks at different heights running at different rates, etc.

(b) A vector whose tangent satisfies $K^a K_a = 0$.

(c) Since $R^\alpha{}_{\beta\gamma\delta} = 0$ in Cartesian coordinates of $\mathbb{R}^{3,1}$, $R = 0$ in Cartesian coordinates of $\mathbb{R}^{3,1}$. But R is coordinate invariant, so $R = 0$ for $\mathbb{R}^{3,1}$ in all coordinates.

(d) Global inertial frames do not exist in gravity: even free falling frames accelerate.

(e) ∂_a does not transform like a vector under Lorentz transformations.

(f) Since $\partial_\mu g_{\nu\rho}|_p = 0$ for L.I.C. at p, this would mean $\partial_\mu g_{\nu\rho} = 0$ in any coordinate system at all p, so $\partial_\mu g_{\nu\rho}$ can't be a tensor.

2

(a)

$$\begin{aligned} R_{abcd}g^{ac} = R_b d &= -\frac{1}{l^2}(g_{ac}g^{ac}g_{bd} - g_{ad}g_{bc}g^{ac}) \\ &= -\frac{1}{l^2}(4g_{bd} - \delta^c_d g_{bc}) \\ &= -\frac{1}{l^2}(4g_{bd} - g_{bd}) = -\frac{3}{l^2}g_{bd}. \end{aligned} \tag{1}$$

(b)

$$R = R_{bd}g^{bd} = -\frac{3}{l^2}g_{bd}g^{bd} = -\frac{12}{l^2} \tag{2}$$

(c)

$$G_{ab} = R_{ab} - \frac{1}{2}Rg_{ab} = -\frac{3}{l^2}g_{ab} + \frac{6}{l^2}g_{ab} = \frac{3}{l^2}g_{ab} \tag{3}$$

(d)

$$\begin{aligned} K &= R_{abcd}R^{abcd} \\ &= \frac{1}{l^4}(g_{ac}g_{bd} - g_{ad}g_{bc})(g^{ac}g^{bd} - g^{ad}g^{bc}) \\ &= \frac{1}{l^4}(16 - g_{ac}g_{bd}g^{ad}g^{bc} - g_{ad}g_{bc}g^{ac}g^{bd} + 16) \\ &= \frac{1}{l^4}(32 - \delta^c_d\delta^d_c - \delta^d_c\delta^c_d) \\ &= \frac{24}{l^4}. \end{aligned} \tag{4}$$

3

(a) This is the curve $(t, r, \theta, \phi) = (t, r_0, \theta_0, \phi_0)$. A particular tangent is

$$U^\mu \to \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}. \tag{5}$$

(b)

$$U^\mu U^\nu g_{\mu\nu} = -\left(1 - \frac{2GM}{r}\right). \tag{6}$$

For $r > R_s$, this is negative, so U^μ is timelike.

(c) It is not true. U^μ is not a 4-velocity because we did not parametriz it by τ: we parametrized it by t.

4

The time delay between clocks at different heights (approximating g = constant between your head and feet):

$$\frac{\Delta\tau_l}{\Delta\tau_0} = e^{gl/c^2}. \tag{7}$$

If you are 20 years old according to your feet and have a height of 1.75m (in your rest frame), then

$$\Delta\tau_l - \Delta\tau_0 = \Delta\tau_0(e^{gl/c^2} - 1) = 1.4 \cdot 10^{-7} s. \tag{8}$$

5

In the police car frame (the primed frame below):

$$v_c' = \frac{v_p - v_c}{1 - v_p v_c} = \frac{2}{5}. \tag{9}$$

The bullets need to be traveling at $v'_{\text{bullet}} > \frac{2}{5}c$.

6

The student neglected to account for time dilation: $t = \gamma\tau_0$, where $\tau_0 = 2 \cdot 10^{-6}$ s (rest lifetime).

$$vt = v\gamma\tau_0 = 2000m. \tag{10}$$

Using the definition of γ, we get

$$\frac{v^2}{1-v^2/c^2} = \left(\frac{2000m}{2\cdot 10^{-6}s}\right)^2, \tag{11}$$

where we have inserted c for dimensional consistency. We find $v = 2.87\cdot 10^8$ m/s.

7

(a) The trajectory is $(\frac{1}{\alpha}\sinh\alpha\tau, \frac{1}{\alpha}\cosh\alpha\tau, 0, 0)$, so the 4-velocity is

$$U^\mu = \frac{dx^\mu}{d\tau} = \begin{bmatrix}\cosh\alpha\tau\\ \sinh\alpha\tau\\ 0\\ 0\end{bmatrix}. \tag{12}$$

(b)

$$\begin{aligned} U^\mu\nabla_\mu U^\nu &= U^\mu(\partial_\mu U^\nu + \Gamma^\nu_{\mu\rho}U^\mu U^\rho)\\ &= U^\mu\partial_\mu U^\nu\\ &= \frac{dx^\mu}{d\tau}\partial_\mu U^\nu = \frac{dU^\nu}{d\tau} \neq 0. \end{aligned} \tag{13}$$

The trajectories do not solve the geodesic equation.

(c) $W^a\nabla_a V^b$

8

(a)

$$r_* = \int \frac{rdr}{r-2GM} = r + 2GM\ln\left(\frac{r}{2GM}-1\right). \tag{14}$$

r_* goes to $-\infty$ as $r\to 2GM$. This is a very slow coordinate!

(b) u = constant along lines moving towards larger r. v = constant along lines moving towards smaller r.

(c)

$$ds^2 = -\left(1-\frac{2GM}{r(u,v)}\right)dudv(+r(u,v)^2d\Omega^2), \tag{15}$$

and $g_{uu} = 0 = g_{vv}$.

(d)

$$r(u,v) + 2GM\ln\left[\frac{r(u,v)}{2GM}-1\right] = \frac{v-u}{2}, \tag{16}$$

from which we get

$$e^{\frac{r}{2M}}\left(\frac{r}{2M}-1\right)=e^{\frac{v-u}{4M}}, \tag{17}$$

so

$$1-\frac{2M}{r}=\frac{2M}{r}. \tag{18}$$

$$ds^2=-\frac{2M}{r}e^{-\frac{r}{2M}}e^{\frac{v-u}{4M}}dudv \tag{19}$$

(e)

$$dU=\frac{1}{4M}e^{-\frac{u}{4M}}du \tag{20}$$

$$dV=\frac{1}{4M}e^{\frac{v}{4M}}dv \tag{21}$$

$$ds^2=-\frac{32(GM)^3}{r}e^{-\frac{r}{2M}}dUdV+r^2d\Omega^2 \tag{22}$$

(f) $r(U,V)>0$
(g)

$$e^{\frac{r}{2M}}\left(\frac{r}{2M}-1\right)=e^{\frac{v-u}{2M}}=-VU \tag{23}$$

(h) When $r=0, UV=1$. $r=0$ occurs twice. When $r\to+\infty, UV\to-\infty$. $U>0, V>0$ and $U<0, V<0$ correspond to two singularities. For $r>0$, $e^{\frac{r}{2MG}}(r/2MG-1)$ ranges from -1 to +m, so the largest possible value of UV is 1. The constraint is $U\in(-\infty,\infty)$ and $V\in(-\infty,\infty)$, but $UV<1$. The left asymptotic region is $U\to-\infty$, and the right asymptotic region is $V\to-\infty$. See Fig (1).

9

(a) In the rest frame of the Higgs, the photons have equal and opposite 3-momentum. The total momentum is

$$p^\mu\to\begin{bmatrix}m_H\\0\\0\\0\end{bmatrix}. \tag{24}$$

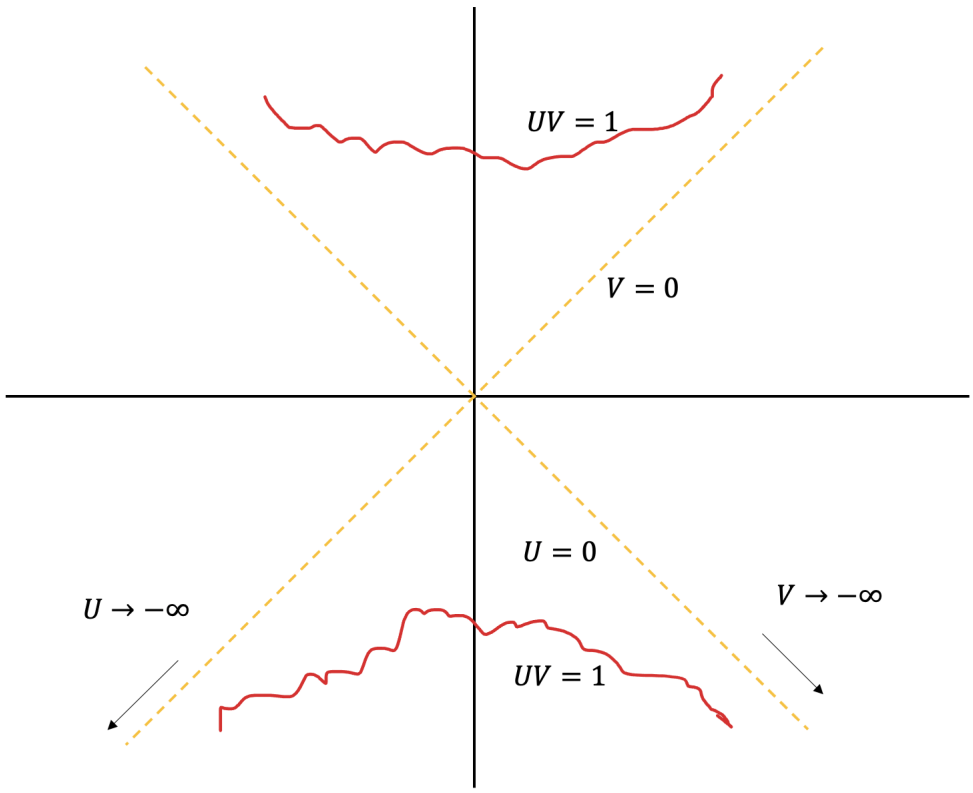

Figure 1: Diagram for problem 8.

This is conserved, so after the collision,

$$p^0 = 2p^0_\gamma = m_H, \tag{25}$$

so $p^0_\gamma = m_H/2$ each.

(b) We have no information about the second photon. Call its momentum $p^a_{\gamma'}$. Call the momentum of the first photon p^a_γ.

$$p^a_{\gamma'} = p^a_H - p^a_\gamma. \tag{26}$$

We also know $p^a_{\gamma'} p_{\gamma' a} = 0$. Altogether

$$\begin{aligned} 0 &= (p^a_H - p^a_\gamma)(p_{Ha} - p_{\gamma a}) = p^a_H p_{Ha} + p^a_\gamma p_{\gamma a} - 2p^a_H p_{\gamma a} \\ &= -m_H^2 + 2E_H E_\gamma - 2p^i_H p_{\gamma i} \\ &= -m_H^2 + 2E_H E_\gamma - 2E_\gamma \cos\theta \sqrt{-m_H^2 + E_H^2}. \end{aligned} \tag{27}$$

We solve for θ and get

$$\theta = \cos^{-1}\left[\frac{2E_H E_\gamma - m_H^2}{2E_\gamma \sqrt{-m_H^2 + E_H^2}}\right]. \tag{28}$$

10

(a)

$$L_\gamma = \int_\gamma \sqrt{ds^2} \tag{29}$$

, where γ is a path. In Euclidean space,

$$L_E = \int_{\theta=0}^{2\pi} \sqrt{dr^2 + r^2 d\theta^2} = \int_0^{2\pi} \sqrt{(dr/d\theta)^2 + r^2} d\theta = r \int_0^{2\pi} d\theta = 2\pi r. \tag{30}$$

The space is called the Lobachevsky space:

$$L_L = \int_{\text{circle}} \sqrt{ds^2} = \int_{v=0}^{2\pi} \sqrt{du^2 + \cosh^2 u dv^2} = \int_{v=0}^{2\pi} \cosh u dv = 2\pi \cosh u. \tag{31}$$

(b)

$$A_E = \int_{\theta=0}^{2\pi} \int_{r=0}^{R} R dr d\theta = \pi R^2 \tag{32}$$

$$A_L = \int_{U=0}^{2\pi} \int_{u=0}^{U} \cosh U du dv = 2\pi \sinh U \tag{33}$$

(c) Euclidean space is flat while Lobachevsky space is not. Lobachevsky space is hyperbolic (kind of like a wormhole). Circumference and area increase much faster with radius!